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DEPARTMENT OF COMPUTER ENGINEERING 2025-2026

CLOUD BASED STUDENT MANAGEMENT SYSTEM

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AIM

To design and develop a scalable Cloud-Based Student Management System using React (Frontend) and Spring Boot (Backend), deployed on modern cloud platforms such as Vercel and Render. The system integrates Supabase as a cloud database, Docker for containerization, DockerHub for image management, and Jenkins with SonarCloud for CI/CD automation.

The objective is to provide secure, real-time, intelligent academic management with Machine Learning-based CGPA prediction.

ABSTRACT

- The Cloud-Based Student Management System is a full-stack web application designed to manage student academic data securely and efficiently.
- The frontend is developed using React.js and deployed on Vercel, providing fast global access through CDN infrastructure. The backend is built using Spring Boot and hosted on Render for reliable cloud execution.
- The system uses Supabase (PostgreSQL) as a managed cloud database to ensure secure, scalable, and centralized data storage.
- The backend application is containerized using Docker, and Docker images are stored on DockerHub for version control and easy deployment.
- A CI/CD pipeline implemented using Jenkins and Sonar Cloud automates code building, testing, quality analysis, and deployment.
- Additionally, a Machine Learning module built with Python and Flask predicts students' CGPA based on academic performance data, providing intelligent insights for academic planning.

LITERATURE SURVEY

- Research studies show that cloud-based systems improve scalability, performance, and availability for large numbers of users.
- DevOps practices and CI/CD pipelines significantly reduce deployment errors and increase system reliability.
- However, many existing systems lack integration between cloud deployment, DevOps automation, and predictive analytics.
- This project combines all these technologies into a single integrated solution.

PROBLEM STATEMENT

- Traditional student management systems suffer from limited scalability, manual deployment processes, and lack of intelligent analytics.
- These issues result in poor system performance under high user load, frequent deployment errors, and limited support for data-driven academic decision-making.
- There is a need for a cloud-based, automated, and intelligent system to efficiently manage academic information.

EXISTING SYSTEM

- The existing student management systems are mostly basic web or desktop applications with limited scalability and performance.
- They rely on manual or semi-automated deployment, which leads to frequent errors and downtime.
- Most systems do not provide advanced analytics or Machine Learning-based performance prediction.

PROPOSED SYSTEM

- It provides more secure, role-based access for students, teachers, and administrators using JWT authentication.
- The system enables real-time management of attendance, marks, and academic records.
- A Machine Learning module is integrated to predict students' CGPA and analyze performance trends.
- DevOps tools are used to automate testing, code quality checks, and deployment.
- This improves system reliability, performance, and ease of maintenance.

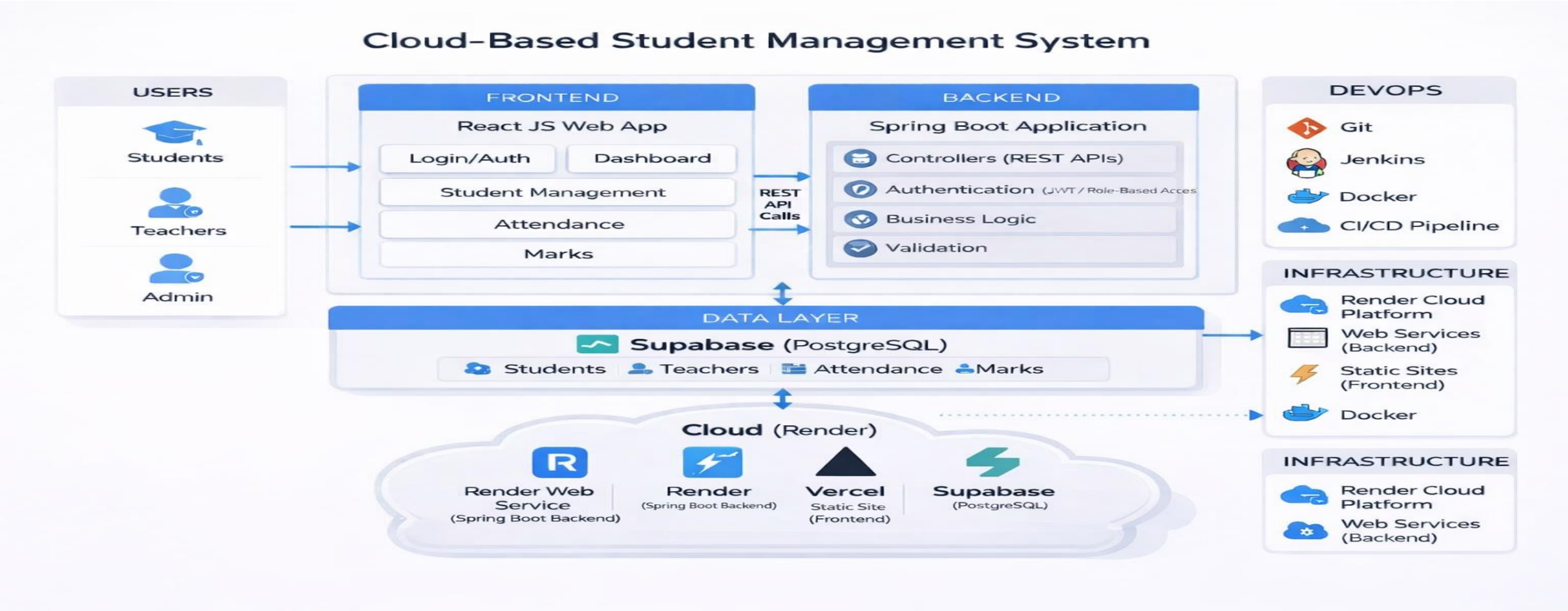
HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or higher (i5 recommended)
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: 256 GB SSD or higher
- Display: Standard monitor (Full HD recommended)
- Internet connection for cloud deployment

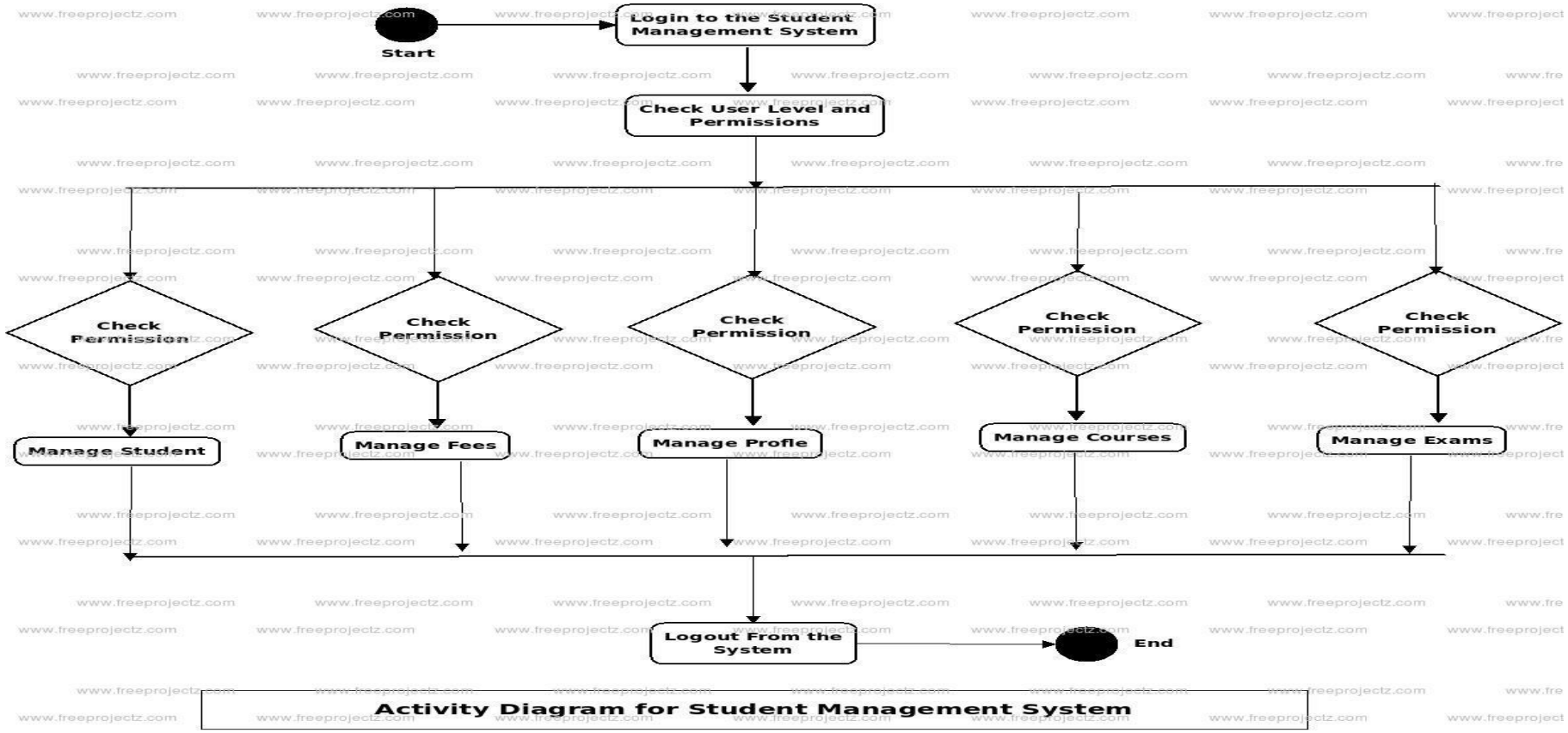
SOFTWARE REQUIREMENTS

Frontend	Backend	Machine Learning	DevOps	Cloud
React js	Spring Boot (Java)	Python	Jenkins	Rendar
Bootstrap	Maven	Flask	SonarCloud	Vercel
Axios	PostgreSQL (Supabase)	scikit-learn, pandas	Docker	DockerHub,Supabase(cloud database)

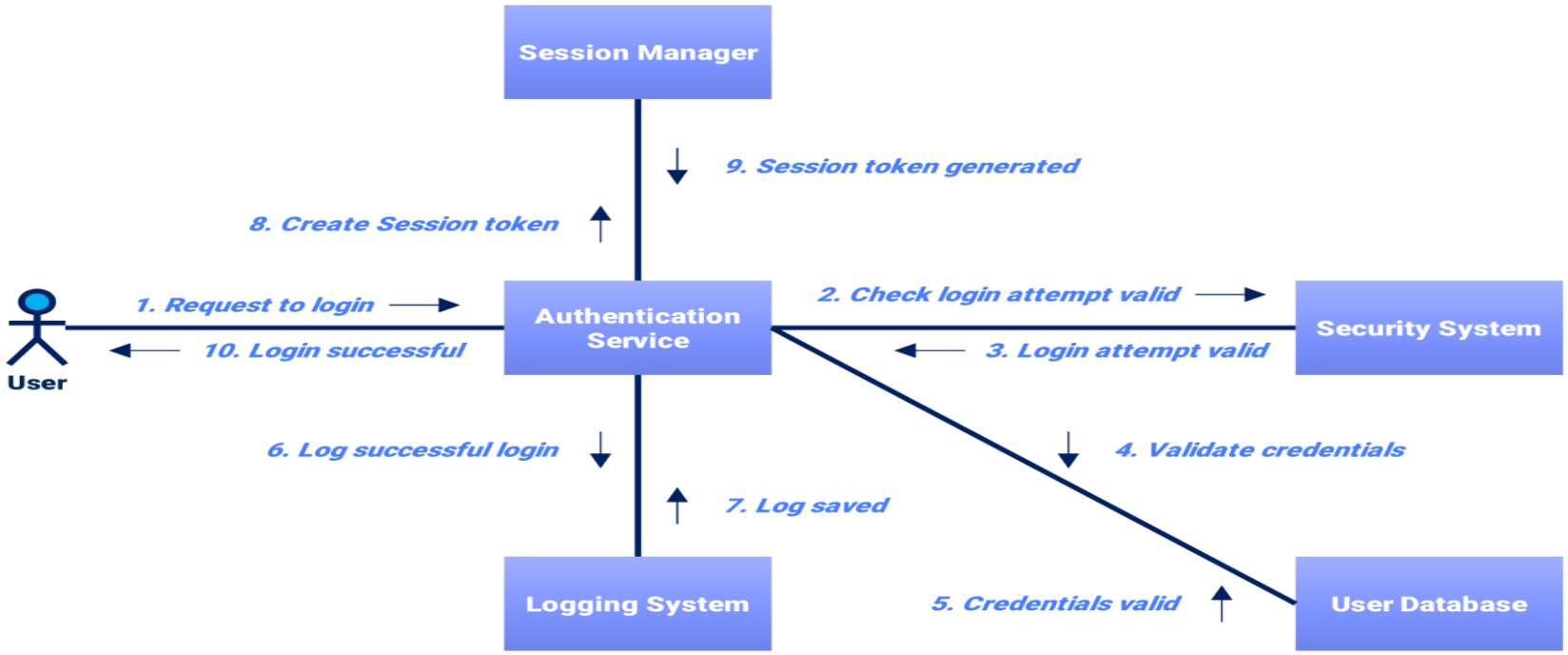
SYSTEM ARCHITECTURE



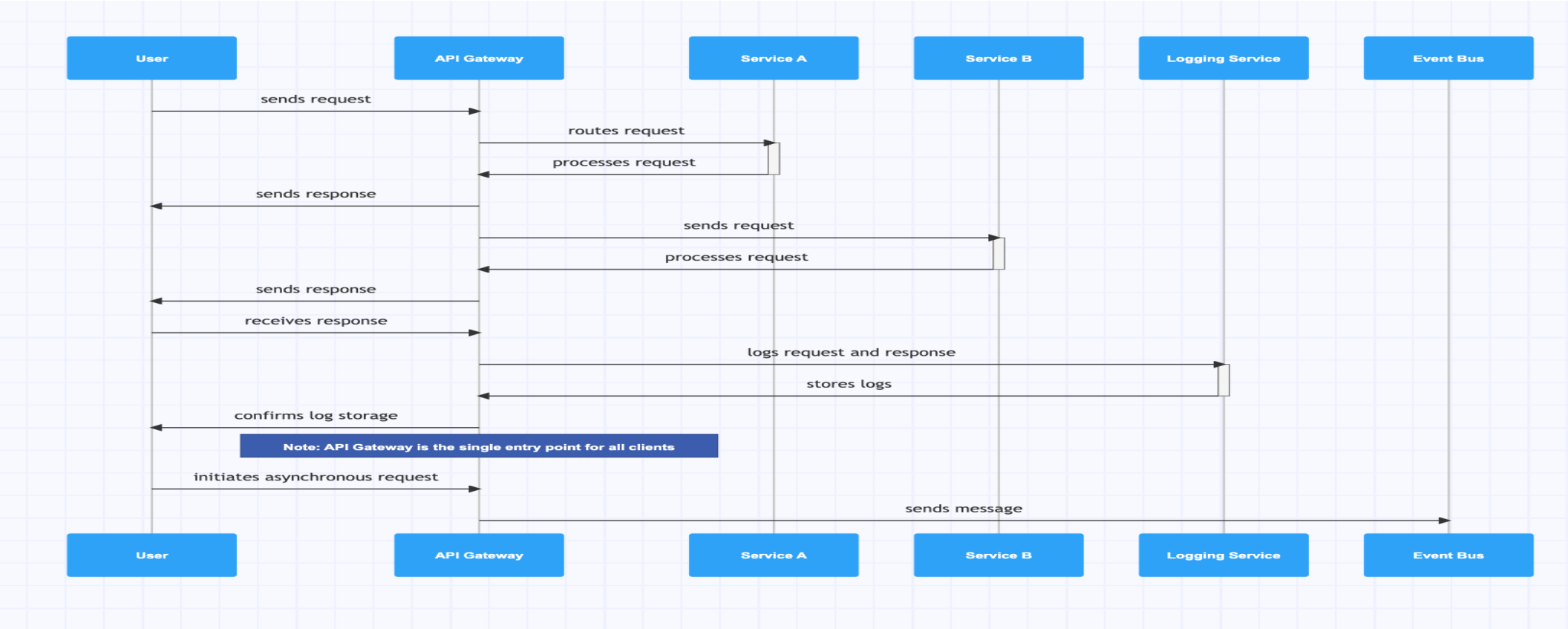
ACTIVITY DIAGRAM



COLLABORATION DIAGRAM



SEQUENCE DIAGRAM



LIST OF MODULES

- Admin Module
- Teacher Module
- Student Module
- Machine Learning Module
- Cloud & Storage Module
- DevOps & CI/CD Module

Admin Module

It allows the administrator to manage students, teachers, and academic details.

The admin can:

- Add, update, and delete student records
- Add and manage teacher details
- Manage subjects and academic years
- View system reports

This module ensures that the system runs smoothly and that all data is properly maintained and secured.

Teacher Module

The Teacher Module enables faculty members to manage academic activities efficiently.

Teachers can:

- Upload and update student marks
- Record and manage attendance
- Monitor student academic progress
- View analytics and performance trends

This module ensures accurate academic record maintenance and improves communication between teachers and students.

Student Module

The Student Module provides secure and personalized access to academic information.

Students can:

- View attendance records
- Check examination marks
- Access CGPA prediction results
- Download academic reports

This module improves transparency, accessibility, and student engagement by providing real-time academic updates.

Machine Learning Module

The Machine Learning Module enhances the system with predictive analytics capabilities.

This module:

- Analyzes historical academic data
- Predicts student CGPA using trained ML models
- Identifies students at academic risk
- Generates performance trend insights

It is implemented using Python and Flask, integrated with the backend via REST APIs.

The model can be retrained periodically to improve accuracy.

Cloud & Storage Module

The Cloud & Storage Module ensures scalability, availability, and secure remote access.

It includes:

- Frontend hosted on Vercel
- Backend deployed on Render
- Supabase (PostgreSQL) as cloud database
- Remote accessibility from anywhere

This module enables high availability, centralized data storage, and improved system scalability.

DevOps & CI/CD Module

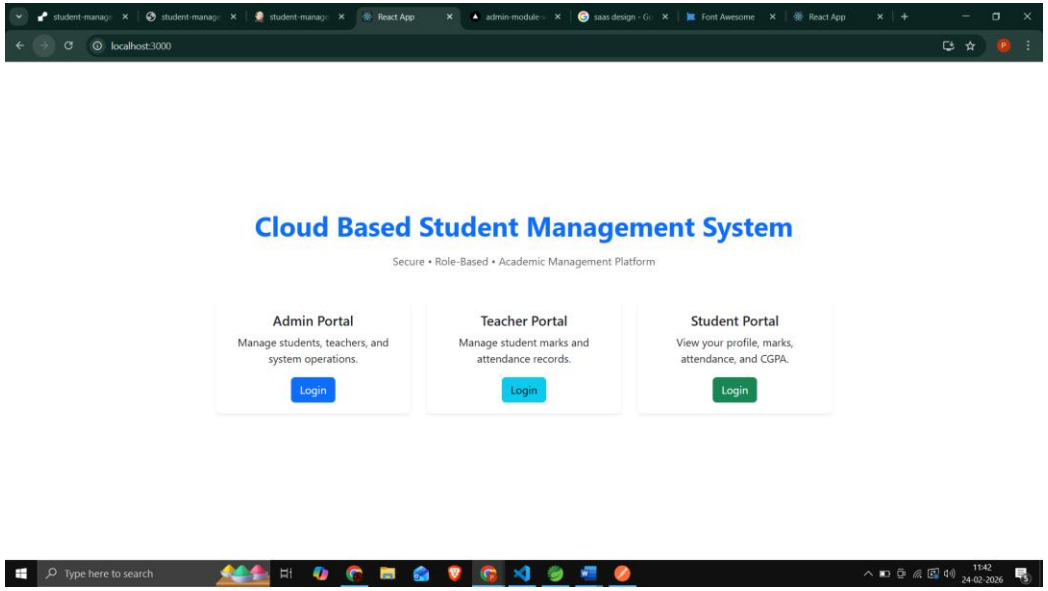
The DevOps & CI/CD Module automates the development and deployment lifecycle.

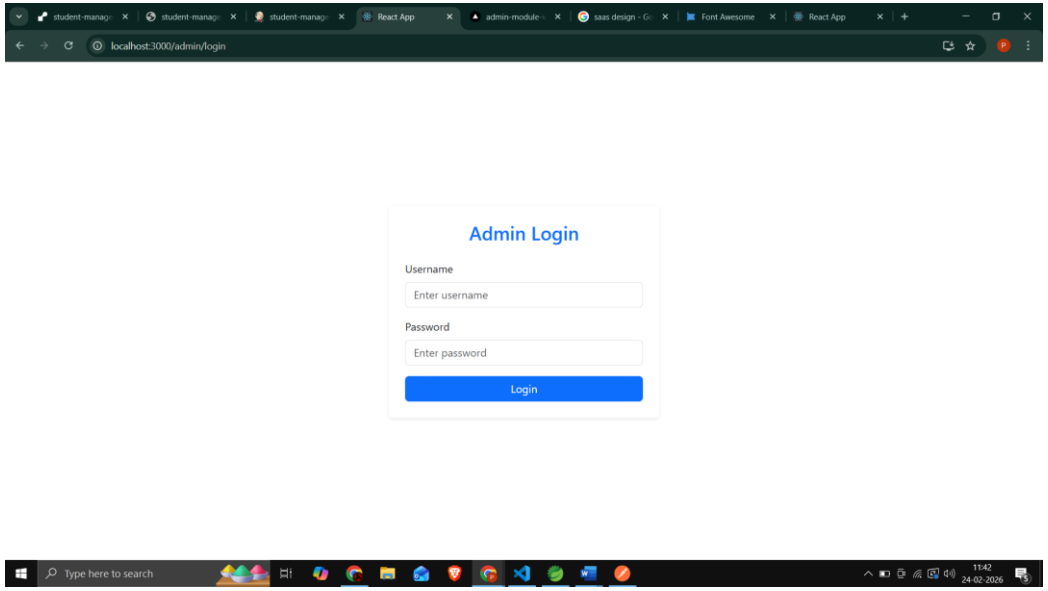
It includes:

- GitHub for version control
- Jenkins for automated build pipeline
- SonarCloud for code quality analysis
- Docker for containerization
- DockerHub for image storage
- Automated deployment to cloud platforms

This module reduces manual errors, improves reliability, and ensures continuous integration and delivery.

ScreenShots





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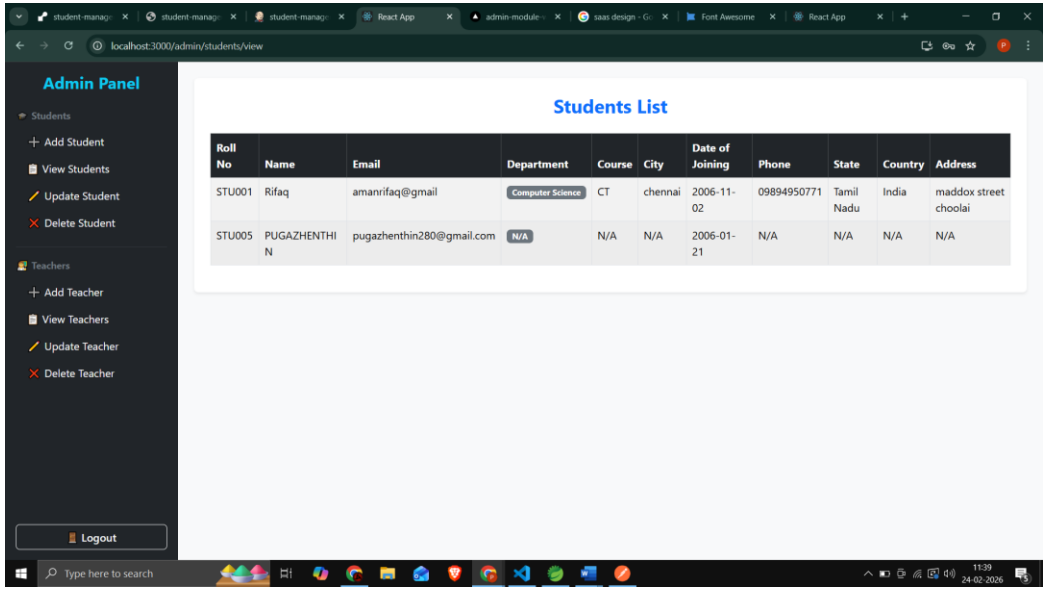

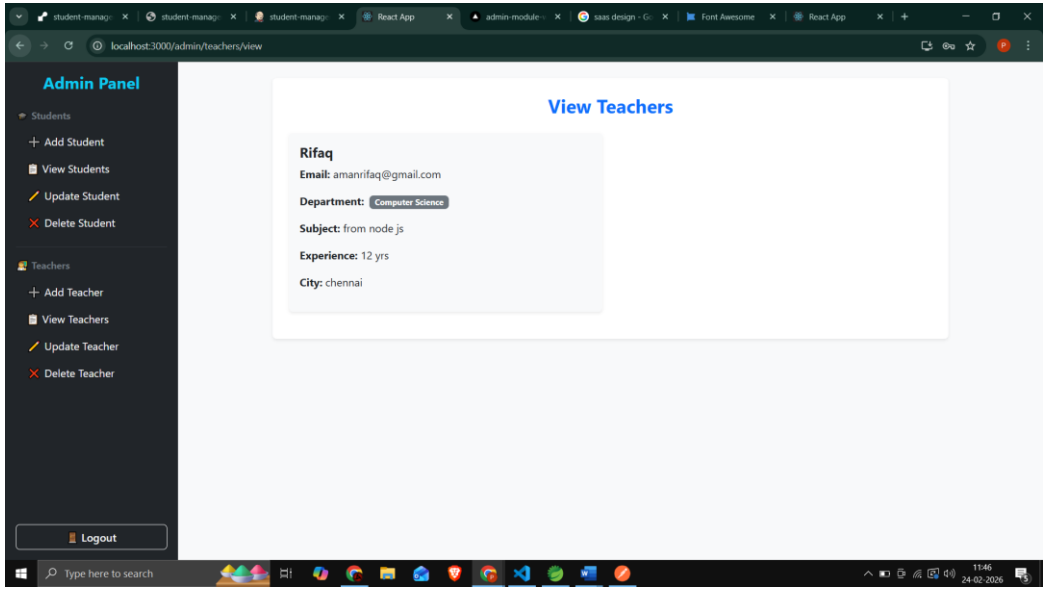

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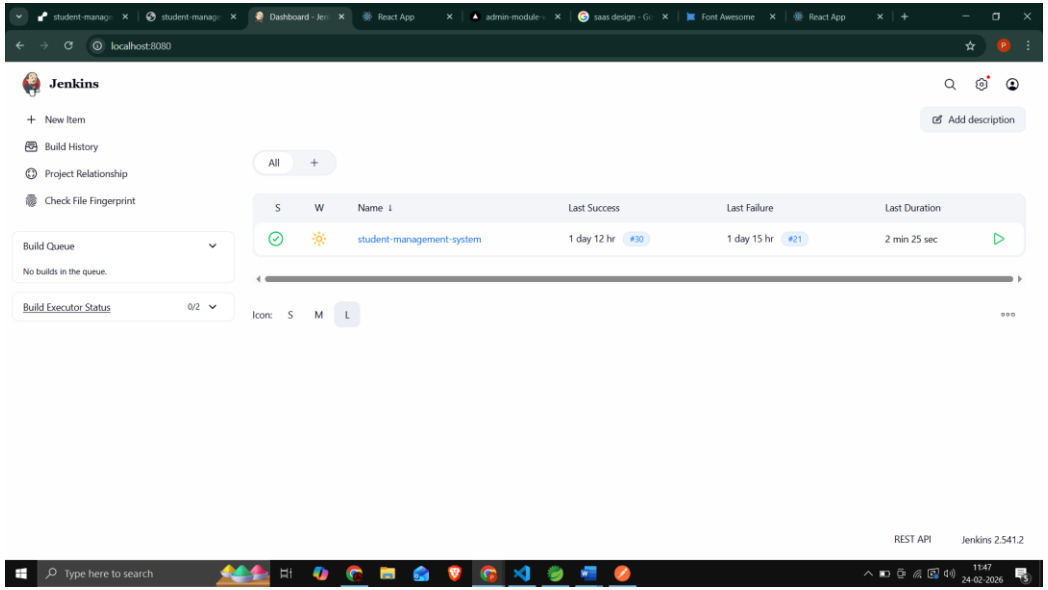

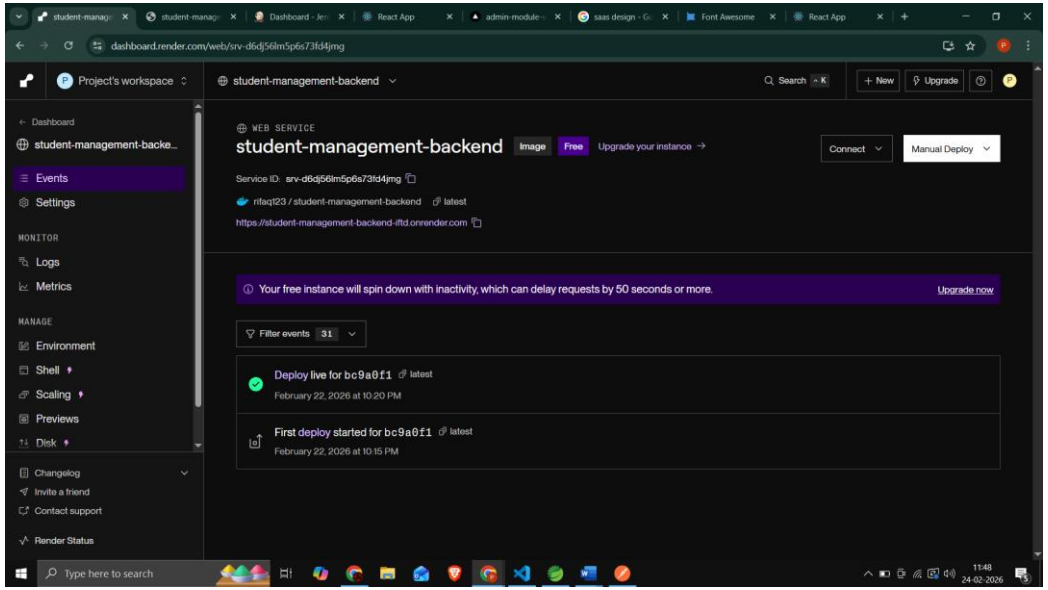

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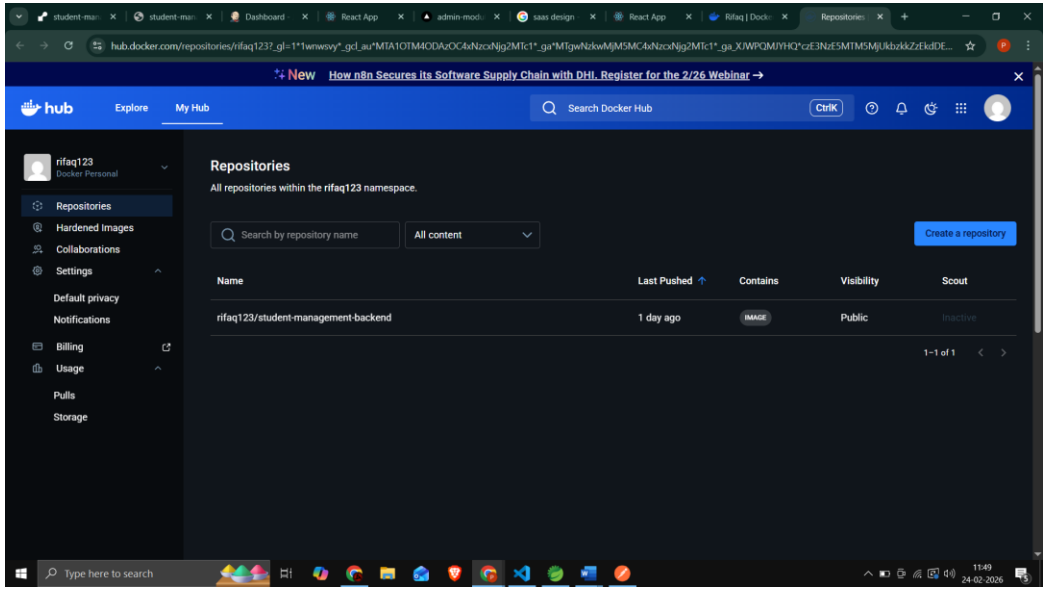

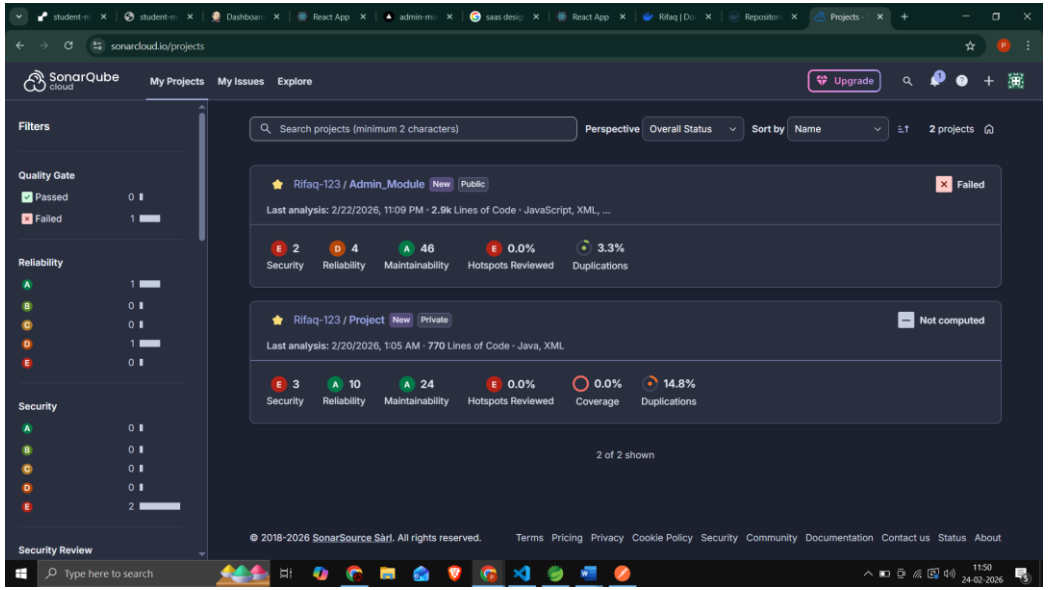

CONCLUSION

- The Cloud-Based Student Management System successfully integrates cloud computing, DevOps automation, and Machine Learning into a single scalable platform.
- By using Vercel for frontend hosting, Render for backend deployment, Supabase for database management, Docker for containerization, and Jenkins with SonarCloud for CI/CD automation, the system ensures reliability, scalability, and security.
- The integration of Machine Learning enhances academic monitoring by providing predictive insights into student performance.
- The system demonstrates how modern cloud technologies can transform traditional academic management into an intelligent, automated platform.

FUTURE ENHANCEMENTS

- Online Examination Module Allow students to attend tests, auto-evaluated quizzes, and generate instant results.
- Chatbot Support Provide AI-based assistance for students to ask queries or get help with navigation.
- Multi-College Support Extend the system to handle multiple institutions under a single platform.
- Mobile Application Support Develop Android and iOS apps to allow students and teachers to access the system from mobile devices.

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THANK YOU